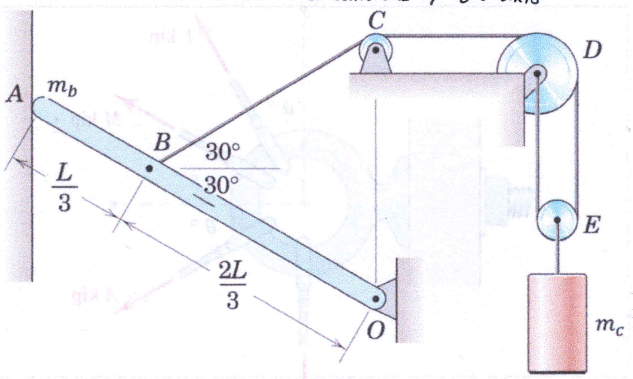
Exam 01 (Preliminary) for ENS161
A.Y. 2026-2027, S1

Name CHRISTIAN CAHIG
Section _____

Instructions. Fill in the information asked above. Submit this questionnaire along with your answer sheets.

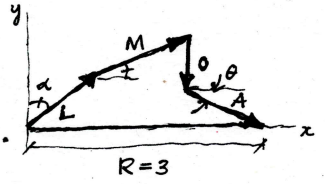
Part I. Write your answers in the space provided after each question. For identification questions, each item required is worth one a point. For true-or-false questions, write *only* T or F. For fill-in-the-blank questions, each blank is worth a point.

1. A static system of forces and/or moments acting on a rigid body is called _____ if the unknown support reactions can be found by using only the equilibrium equations.
   determinate / statically determinate
2. An object is idealized as a ___ if its dimensions are deemed negligible.
   particle
3. *True or False.* If a rigid body is in static equilibrium, then the total moment due to the forces (including support reactions) about a point outside the body is zero.
   T
4. Two cables sustain 12- and 14-lb tensions when loaded with a 1-tonne mass. What would be the cable tensions if the mass is halved?
   6.7 lb
5. An object at rest may be analyzed as a particle even if its dimensions are not negligible, provided that the forces acting on it are ___.
   concurrent
6. Two systems of forces and moments cause the same net force and net moment on an object. The systems are said to be ___.
   equivalent
7. *True or False.* Two forces having equal magnitudes but opposing directions form a couple.
   F
8. With this theorem, the resultant moment of a system of forces about an axis is the algebraic sum of the moments of each of the forces about the same axis.
   Varignon's theorem / principle of moments
9. *True or False.* Concurrent forces are not always coplanar.
   T
10. The principle of transmissibility of forces effectively treats a force as a ____ vector.
    sliding
11. *True or False.* A statically determinate system is always stable.
    F
12. *True or False.* The moment of a force $\boldsymbol{F}$ about a point A is perpendicular to both $\boldsymbol{F}$ and the position vector from A to any point in the line of action of $\boldsymbol{F}$.
    T
13. A force vector is characterized by its ___, ___, and ___.
    magnitude
    direction
    point of application
14. Identify three strategies to maximize the moment of a force $\boldsymbol{F}$ about a point A.
    maximize magnitude of $\vec{F}$
    maximize moment arm (perpendicular distance) from $\vec{F}$ to A
    make position vector from A to any point of interest along $\vec{F}$, and $\vec{F}$ as orthogonal as possible
15. Identify three scenarios for which a force does not cause a rotational tendency about an axis.
    that force intersects the axis at some point
    that force runs parallel to the axis
    the line of action of the force coincides w/ the axis



16. Identify two scenarios wherein the bar is a three-force member. m_b and m_c are the respective masses of the bar and of the cylinder.
    $m_c = 0$ (The bar is subject to its weight, the contact force at A, and the reaction by the pin at O.)
    $m_c = \frac{3}{4} m_b$ (The bar is subject to its weight, the cable tension, and the reaction by the pin at O. The cylinder weight just balances the bar weight so that the bar "touches but does not rest upon" the wall at A.)

Problem 1. Let R be the magnitude of the resultant of the four concurrent forces. The limiting values of M and θ correspond to when $R = 3$.



By geometry,

$$\begin{cases} 2\sin(30°) + M\cos(30°) + 1.2\cos\theta = 3 & [1] \\ 2\cos(30°) + M\sin(30°) - 1.8 - 1.2\sin\theta = 0 & [2] \end{cases}$$

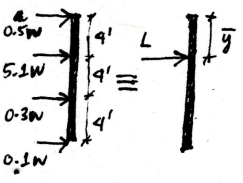
By some algebra,

[redacted]

$$\frac{M\sin(30°)}{M\cos(30°)} = \frac{1.8 - \sqrt{3} + 1.2\sin\theta}{3 - 1 - 1.2\cos\theta} = \tan(30°) \Rightarrow \theta \cong 21.65554234°$$

$$\Rightarrow M \cong 1.021560069 \text{ kips}$$

Problem 2



Because the two force systems are equivalent, their net forces and net moments (say, about D) must be equal. So,

- $L = 0.5w + 5.1w + 0.3w + 0.1w = 6w$ lb
- $L\bar{y} = 6w\bar{y} = (5.1w)(4) + (0.3w)(8) + (0.1w)(12)$

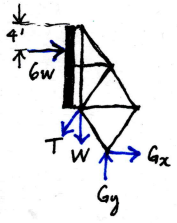
$\Rightarrow \boxed{\bar{y} = 4 \text{ ft}}$

The signboard weighs $(3 \text{ lb/ft})(12 \text{ ft}) = 36$ lb. The tower weighs

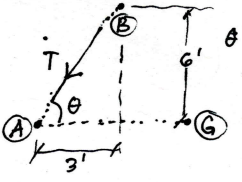
$$(2 \text{ lb/ft})\left[(3 \times 6\text{ft}) + 3\text{ft} + (5 \times 3\sqrt{5}\text{ ft})\right] = 42 + 30\sqrt{5} \text{ lb} \cong 109.0820393 \text{ lb}$$

Thus, the combined weight of the signboard and the tower is

$$W = 78 + 30\sqrt{5} \text{ lb} \cong 145.0820393 \text{ lb}$$



Let T be the tension in cable AB. Also observe that:



$\theta = \tan^{-1} 2 \cong 63.43494882°$

Balancing moments about G,

$$T(6)\sin\theta + W(3) = 6w(18-4) \Rightarrow \boxed{T = \frac{28w - 78 - 30\sqrt{5}}{2\sin\theta}} \quad [1]$$

Equating the LHS of [1] to 67 lb, the maximum allowable value of w is just the corresponding value of w:

$$67 = \frac{28\bar{w} - 78 - 30\sqrt{5}}{2\sin\theta} \Rightarrow \boxed{\bar{w} \cong 9.46197439}$$

The values of G_x and G_y corresponding to $\bar{w}$ are obtained by balance of horizontal and of vertical forces, respectively.

$$\bar{G_x} + 6\bar{w} = \underbrace{T}_{67}\cos\theta \Rightarrow \boxed{\bar{G_x} \cong -26.80853544 \text{ lb}}$$

$$\bar{G_y} = \underbrace{W}_{78+30\sqrt{5}} + \underbrace{T}\sin\theta \Rightarrow \boxed{\bar{G_y} \cong 205.0086611 \text{ lb}}$$

Problem 3. Realize that "for every kilogram of the model load" is numerically equal to setting $m = 1$.

(a) $\vec{M}_C = \langle 0, -(1)g(3-1), 0\rangle \cong \langle 0, -19.6133, 0\rangle$ N·m per 1 kg of model load

Alternatively, one could go the "textbook" way: $\vec{r}_{CD} \times 1g\langle 0,0,-1\rangle$

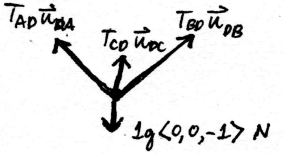
(b) $\vec{M}_{AB} = \langle 0, 1g(1), 0\rangle \cong \langle 0, 9.80665, 0\rangle$ N·m per 1 kg of model load

Alternatively, one could go the "textbook" way:

$$\left[\left(\vec{r}_{BD} \times 1g\langle 0,0,-1\rangle\right) \cdot \vec{u}_{AB}\right] \vec{u}_{AB}$$

(c) By inspection,

- $\vec{r}_{DA} = \langle -1, -4, 6\rangle$ m $\Rightarrow \vec{u}_{DA} = \frac{1}{\sqrt{53}}\langle -1, -4, 6\rangle$
- $\vec{r}_{DC} = \langle 2, 0, 6\rangle$ m $\Rightarrow \vec{u}_{DC} = \frac{1}{\sqrt{10}}\langle 1, 0, 3\rangle$
- $\vec{r}_{DB} = \langle -1, 4, 6\rangle$ m $\Rightarrow \vec{u}_{DB} = \frac{1}{\sqrt{53}}\langle -1, 4, 6\rangle$



Invoking force balance at D,

$$T_{AD}\vec{u}_{DA} + T_{CD}\vec{u}_{DC} + T_{BD}\vec{u}_{DB} + \underbrace{1g}_{m}\langle 0,0,-1\rangle = \langle 0,0,0\rangle$$

or,

$$\begin{cases} -\frac{1}{\sqrt{53}}T_{AD} + \frac{1}{\sqrt{10}}T_{CD} - \frac{1}{\sqrt{53}}T_{BD} = 0 \\ -\frac{4}{\sqrt{53}}T_{AD} + 0\,T_{CD} + \frac{4}{\sqrt{53}}T_{BD} = 0 \\ -\frac{6}{\sqrt{53}}T_{AD} + \frac{3}{\sqrt{10}}T_{CD} + \frac{6}{\sqrt{53}}T_{BD} = \widehat{1g}^{\,m} \end{cases}$$

which gives

$T_{AD} \cong 3.96630498$ N per 1 kg of model load
$T_{CD} \cong 3.44570558$ N per 1 kg of model load
$T_{BD} \cong 3.96630498$ N per 1 kg of model load

(d) The results in (c) allows us to solve for the maximum allowable mass for each cable by

(tension per 1 kg model load)($\overline{m}$ kg model load) = tension limit

Hence, for cables AD and BD:

$3.96630498\,\overline{m} = 76 \Rightarrow \overline{m} \cong 19.161411030454\,5$ kg

while for cable CD:

$3.44570558\,\overline{m} = 67 \Rightarrow \overline{m} \cong 19.444935740699\,7$ kg

for the entire system, the maximum allowable mass is the lesser value of $\overline{m}$, i.e., 19.161411030454 5 kg.

(e) Then the tensions due to the max. allowable mass are:

- AD: $(3.96630498\ N/kg)(19.161411030454\,5\ kg) \cong$ 76 N (AD)
- CD: $(3.44570558\ N/kg)(19.161411030454\,5\ kg) \cong$ 66.0246809 N (CD)
- BD: $(3.96630498\ N/kg)(19.161411030454\,5\ kg) =$ 76 N. (BD)